

The Flaw of Averages

CE 310 · Week 1

Probability & Statistics in Civil and Architectural Engineering

University of Arizona · Fall 2026

Why This Course Matters

Personal story

In the Age of AI

Importance of Probability and Statistics

I studied probability and statistics from high school onward — and for years I did not see why they mattered.

What I eventually understood is that this subject is a **lens**: it is how engineers look at an uncertain world and still make defensible decisions. That is why it is a required course.

You may build a very different career from mine. It will still come down to the same thing — the quality of the decisions you make with incomplete information.

Are you afraid that Artificial Intelligence (AI) will take over your jobs?

A new disruptive technology always arrives. The internet did. Large language models are doing it now.

What does not change is the fundamentals. There are only two places to stand when a technology arrives: **working with it**, or **replaced by it**. Which side you end up on is decided by what you understand.

That is the real reason this course is required. It teaches the part that does not go out of date.

You are actually in a good position.

AI has moved fastest in the digital world — text, images, code. **Physical AI** — systems that act on real buildings, roads and structures — is moving far more slowly.

Civil and architectural engineering is a field of physical assets. That gives you a longer runway than most professions. It will not last forever — Tesla is a preview of what integration looks like — but it is real, and it is yours to use.

AI Tools Are Already Part of Engineering

As an engineer, you might have used:

- Tools that **predict** how much energy a building will use next month
- Tools that **detect** when a sensor or structural component is behaving abnormally
- Tools that **estimate** cost ranges and construction schedules under uncertainty
- Tools that **recommend** design alternatives based on performance data

All of these tools take data and produce a prediction or a decision.

The question is: how do they actually work?

Probability and Statistics: The Foundation of All AI

Every AI tool learns from data. The math it uses to describe patterns, measure uncertainty, and make predictions is **probability and statistics** — nothing else.

What the AI tool does	The math underneath
Predicts tomorrow's building energy use	Fits a curve to past energy and weather data
Flags a sensor reading as suspicious	Checks whether the value falls outside the normal range
Estimates how uncertain a cost forecast is	Simulates thousands of possible outcomes
Predicts the probability of a storm event	Computes likelihood from historical weather records

The tools get fancier. The foundation stays the same.

Why You Need to Understand It — Not Just Use It

You can use a calculator without understanding arithmetic. But you can't trust the answer without knowing how to check it.

The same logic applies to AI:

- AI tools can be wrong — they fail silently when conditions change
- A model trained on 2023 Tucson weather does not know what 2050 will look like
- An algorithm that worked in Phoenix may not apply to Tucson

Who catches the error? The engineer who understands what the tool is actually computing — and that is what this course teaches you to be.

How This Course Is Structured

Each week:

Class 1 (Monday) — Theory lecture
Concepts + worked examples

Class 2 (Wednesday) — Tutorials for
your in-class exercises · PreLab Quiz

Class 3 (Friday) — In-Class Exercises
· Real dataset · Work in class;
finish at home (including your assignment)

Assessments: weekly lab sheets · midterm exam (Week 8) · final exam (Week 15) · capstone project (Week 16)

Lab tools by week:

Weeks 1–8 · Excel

Descriptive stats · Distributions
Percentiles · Conditional statistics

Weeks 9–15 · Python / Colab

Regression · Hypothesis testing
Sensitivity · Monte Carlo · Bootstrap
Capstone project

Why Excel? — Weeks 1–8

It is universally used in engineering practice

Structural calculations, energy models, cost estimates, construction schedules — Excel is on every engineer's desktop, in every firm, at every agency. Knowing it well is a baseline expectation.

It is a gateway to programming thinking

Excel formulas already are programming:

- A function takes inputs and returns an output
- IF is conditional logic
- Filling a formula down a column is a loop

You are learning to think like a programmer — without the syntax burden. When you later write code in Python, the logic will feel familiar.

Why Python? — Weeks 9–15

It is the language of the AI era

It has become the shared language of data scientists, researchers, and engineers worldwide. I want you to be able to use it before you graduate. It is no longer optional.

It is designed to read like plain English

A single script can load a dataset, run a full statistical analysis, produce publication-quality charts, and build a prediction model — all in one place. Free, open source, and runs on any computer.

It has countless libraries

Statistics, optimization, mapping, image analysis, machine learning — there is a mature, free library for almost all of it. Excel has nothing comparable.

Excel → Python: The Path This Course Takes

	Excel	Python
When	Weeks 1–8	Weeks 9–15
You see	Every cell, every formula	Every line of logic
Best for	Building intuition, checking by hand	Large datasets, automation, AI
In practice	Daily tool in every engineering firm	The language of data science and AI

Excel teaches you *what* to compute and *why*. Python teaches you to do it at scale — the way real AI systems do.

You Already Know How to Program — MATLAB → Python

You've already taken the required MATLAB course. The core ideas transfer directly — you are not starting from zero in Week 9.

	MATLAB (prerequisite)	Python (Weeks 9–15)
Core ideas	Loops, conditionals, functions, vectors/matrices	Same ideas — different syntax
You already know	How to break a problem into logical steps	How to translate that logic into a new language
What's new	—	Reading real data files, statistics libraries, plotting library

Python isn't a new way of thinking for you — it's a new dialect for logic you already have.

Python tutorial materials are on D2L from day one — no need to wait.

The Capstone Project — Weeks 15–16

Apply every tool from this course to a real engineering dataset — and deliver a professional analysis.

What you'll do:

- Work in teams of 3 (instructor-assigned); your team chooses its own real-world dataset — any civil/architectural engineering topic, any public source
- Apply the full CE 310 pipeline to a real engineering dataset to reveal meaningful findings; deliver results as a professional client memo

What you'll deliver:

- **Lab notebook** — individual analysis in Python
- **Written report** — team summary for a client
- **Presentation** — short team delivery (~4 min)

💡 The capstone is the payoff for everything you build this semester — every week's tool shows up as a real decision, not an exercise.

Your Instructor

Wooyoung Jung, Ph.D.

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Education

Degree	Field	Institution	Year
Ph.D.	Civil Engineering	Virginia Tech	2020
M.S.	Architectural Engineering	Yonsei University, South Korea	2016
B.S.	Architectural Engineering	Yonsei University, South Korea	2011

Office hours

By appointment — email me.

What My Research Is About

Human-AI-Building Collaboration

My research lab studies how humans, AI, and buildings can work together to make buildings smarter, healthier, and more energy efficient.

- How can humans seamlessly and effectively interact with buildings?
- How can AI make those interactions easier?
- What are the capabilities needed from each entity?
- Physical AI in buildings

💡 Every project in my lab starts with data — sensor readings, energy logs, weather records. The analysis always begins with exactly what this course teaches.

Interested in Building + AI?

Consider Joining my Human Building Synergy (HUBS) Lab

How it usually works: sit in with the lab for a semester, then register for research credit in later semesters (up to three credits) if it is a good fit for both sides.

What we work on: Building AI — how AI can work well with buildings *and* with the people in them.

Who I am looking for: students who are genuinely curious about this area and will follow through on what they commit to. Prior research experience is not required.

Part 1

The Flaw of Averages

A Famous Engineering Failure

In the 1950s, the U.S. Air Force measured **10 physical dimensions** on **4,063 airmen** — height, arm reach, hip width, and seven others.

They designed cockpits to fit the pilot with average measurements on all 10 dimensions.



F-86 Sabre cockpit, the fighter-jet era of the 1950 Air Force study. U.S. Air Force / National Museum of the USAF — public domain.

How many of the 4,063 pilots fit the "average" cockpit?

Zero.

Not one pilot was average on all 10 dimensions simultaneously.

Designing for the average meant the cockpit fit nobody.

The fix — adjustable seats — is now so universal we don't even notice it.

Engineers Make the Same Mistake

Design problem	Average-based approach	What you actually need
HVAC cooling capacity	Size for ~72 °F annual average	Size for the 1% hottest hours
Structural wind load	Design for ~7 mph mean wind	Design for the peak gust pressure
Water management	Base on average rainfall	Base on peak storm intensity
Bridge traffic load	Use average daily truck count	Use the heaviest truck combination

⚠ In every case, a system sized for the average fails when it is stressed most — exactly when it must not fail.

Your Turn — Add a 5th Row

2 minutes · discuss with your neighbor

Think of a case — from civil or architectural engineering — where a system is commonly sized for the average but should be sized for an extreme.

Be ready to share:

- What is the **system**?
- What is the **average-based mistake**, and what should the **design value** actually be?

✓ Your Turn — Sample Answers

Any of these — or your own — is correct

Design problem	Average-based mistake	What you actually need
Stormwater pipe sizing	Size for mean annual rainfall rate	Size for the 25-year design storm intensity
Electrical service panel	Use average household consumption	Use the simultaneous peak demand hour
Foundation bearing	Use average dead + live load	Use peak load combination per ASCE 7
Glazing U-value	Use mean winter temperature	Use ASHRAE 99.6% design cold night

The pattern is the same every time: failure happens at extremes, not averages.

Part 2

Descriptive Statistics

Three questions every engineer asks about data

Three Questions to Ask About Any Dataset

When you look at data for the first time, you need to answer three questions:

1. Center

What is the typical value?

Where does the data cluster?

2. Spread

How much do values vary?

How far do they scatter from the center?

3. Shape

Is the data symmetric?

Are there extremes that matter?

💡 These three questions are the foundation of statistics — and the first step of every AI analysis pipeline.

Predict Before You See the Data

75 seconds · write your guess

We are about to open a dataset of every hour of Tucson weather in 2023 — 8,760 rows.

Before we do: write down your best guess for each of these.

- What is the **annual average temperature** in Tucson?
- What was the **hottest temperature** recorded in 2023?
- How many hours per year do you think **exceeded 100 °F**?

The Dataset — 8,760 Hours of Tucson Weather

Source: NOAA Automated Surface Observing System (ASOS), Tucson International Airport, 2023

Column	What it measures
DateTime	Timestamp — one row per hour
Temp_F	Dry-bulb air temperature (°F)
Wind_Speed_mph	Sustained wind speed (mph)
Relative_Humidity_pct	Relative humidity (%)

- **8,760 rows** = 365 days × 24 hours — every hour of an entire year
- This is a real engineering dataset; the kind of data you will encounter in practice

Center — What the Mean Tells You (and What It Hides)

The **mean** (arithmetic average) is the most common measure of center. It answers: *if all values were equal, what would that value be?*

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where x_i = each data value, $\bar{x}$ = sample mean, n = sample size

The mean has a critical weakness: **it treats every observation equally** — a heat wave hour and a mild spring morning count the same.

- Tucson 2023 annual mean temperature: **$\approx 72^\circ\text{F}$**
- Tucson also recorded **112°F**
- A system sized for 72°F is undersized by 40°F on the hottest day

The engineering question: does the mean matter here — or the extreme?

Center — The Median

The **median** is the middle value when observations are sorted — half fall below it, half above.

Why it matters: the median is not affected by how extreme the extremes are.

$$\tilde{x} = \begin{cases} x_{(\frac{n+1}{2})} & n \text{ is odd} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} & n \text{ is even} \end{cases}$$

where $x_{(k)}$ = the k -th value in sorted order, n = sample size

	Mean	Median
Sensitive to extremes	Yes	No
Best when	Distribution is symmetric	Data is skewed or has outliers

Example — construction cost overruns:

A few catastrophic projects can pull the mean far above what a typical project costs. The median tells you what a *typical* project actually cost.

Calculating the Median — Step by Step

Odd n — 5 values (°F)

Sort: **28, 58, 73, 86, 112**

$$\text{Position: } \frac{n + 1}{2} = \frac{5 + 1}{2} = 3$$

Median = **73 °F** (3rd value)

💡 The single middle value — no averaging needed.

Even n — 4 values (°F)

Sort: **58, 65, 86, 112**

Middle two: positions 2 and 3

$$\text{Median} = \frac{65 + 86}{2} = \mathbf{75.5\text{ }^{\circ}\text{F}}$$

💡 Average the two middle values when n is even.

Key property: change 112 °F to 500 °F — the mean jumps by 78 °F, the median stays at 73 °F. Position determines the median; magnitude does not.

Center — Mode

The **mode** is the most frequently occurring value in a dataset.

For discrete data — mode is directly useful:

- Most common storm count per month
- Most frequent wind direction
- Most common inspection rating

For continuous data like temperature — raw values rarely repeat exactly, so we bin the data first (e.g., 70–75 °F range) and ask: which bin has the most hours?

Three measures of center — summary:

Measure	What it answers
Mean	If all values were equal, what would they be?
Median	What is the middle value?
Mode	What value appears most often?

For symmetric distributions: mean $\approx$ median $\approx$ mode.

When they diverge, the distribution is **skewed** — more on this shortly.

Spread — Standard Deviation

Standard deviation (s) — the typical distance any observation sits from the mean.

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where x_i = each data value, $\bar{x}$ = sample mean, n = sample size

Tucson 2023 outdoor temperature:

- $s = 18\text{ °F}$, mean = 72 °F
- A typical hour sits 18 °F from the annual mean
- Typical range: roughly $54\text{--}90\text{ °F}$

Excel formula:

`=STDEV( range )` — divides by $n - 1$ (Bessel's correction, covered on the previous slide)

Use `STDEV` (sample) for almost every real engineering dataset — you have a sample, not the full population.

Spread — Coefficient of Variation (CV)

Coefficient of variation (CV) = $s \div \text{mean}$ — expresses spread as a fraction of the mean, removing units so you can compare across different variables.

$$CV = \frac{s}{\bar{x}} \times 100\%$$

	s	Mean	CV
Outdoor temp	18 °F	72 °F	25%
Zone temp (<i>Week 5</i>)	2.9 °F	73.5 °F	4%

Same building, same city — very different spread relative to the mean.

Reading CV as a design signal:

- **CV < 10%:** data clusters tightly around the mean — mean is a reliable design value
- **CV 10–20%:** moderate spread — mean needs context; report s alongside it
- **CV > 20%:** high spread — the mean is likely misleading as a standalone design value; use percentiles

⚠ Outdoor temperature: CV = 25%. The mean (72 °F) describes only the center. The system must handle 108 °F peaks — 36 °F above that mean.

Small vs. Large CV — What It Looks Like

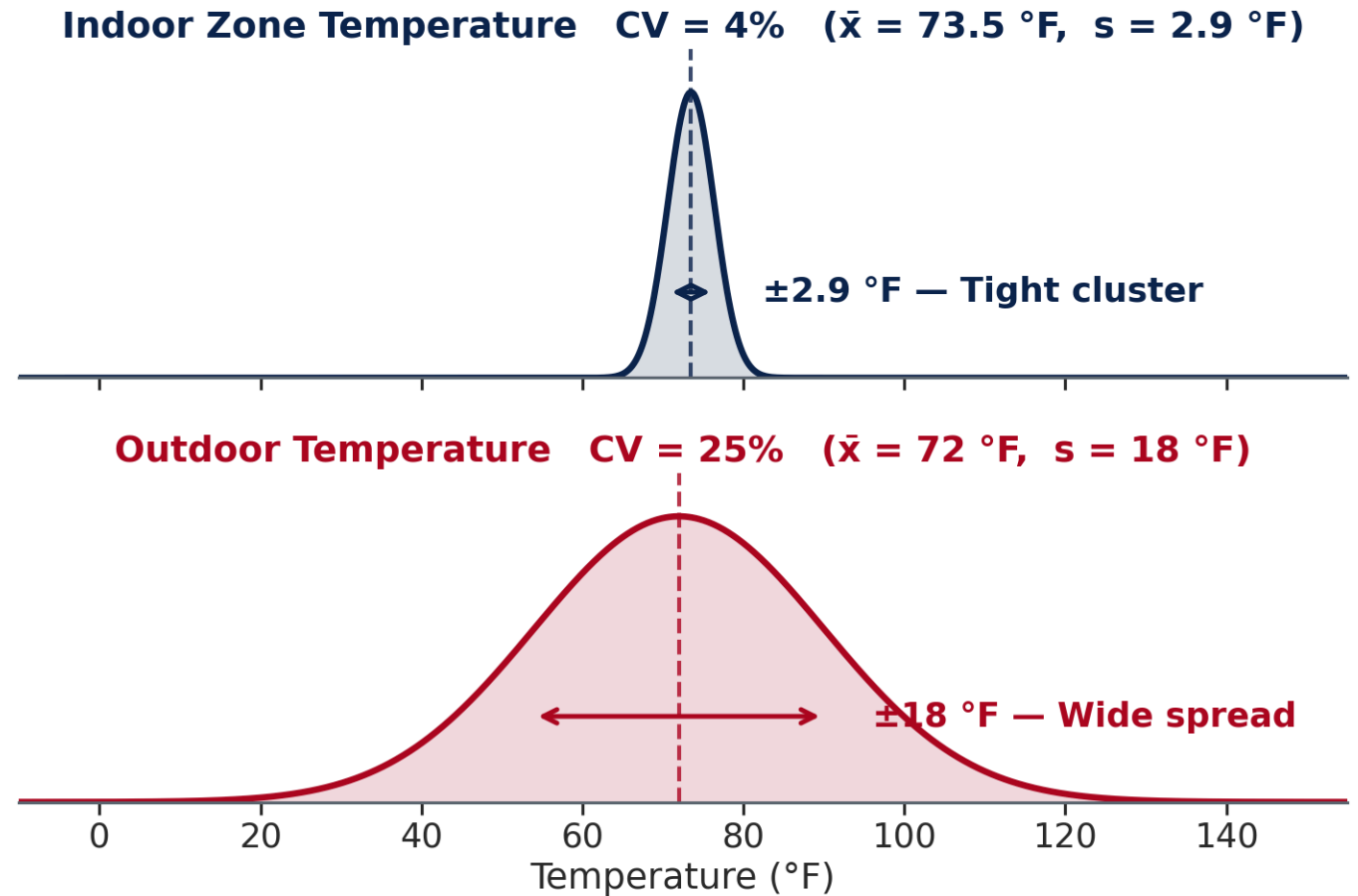
CV = 4% → mean is reliable

Indoor zone temperatures cluster tightly around the setpoint. The mean describes nearly every hour well.

CV = 25% → use percentiles

Outdoor temperatures spread widely across the year. The mean sits at the center but misses the extremes that drive equipment sizing.

Same city, same building — very different spread relative to the mean.



Sample vs. Population — What's the Difference?

Population — every possible value that could ever exist for what you're studying (all Tucson temperatures, past and future).

Sample — the actual data you have in hand right now (2023's 8,760 recorded hours).

💡 You almost never have the full population — only a sample drawn from it. Every statistic you compute from a sample (mean, SD, quartiles) is an **estimate** of the true population value, not the value itself.

Why this distinction matters: the spread formula changes depending on which one you have — next slide.

Sample vs. Population Standard Deviation

	Formula	When to use
Population σ	Divide by n	You have every value (rare)
Sample s	Divide by $n-1$	You have a subset (almost always)

Why $n-1$? A sample tends to underestimate the true population variability. Dividing by $n-1$ instead of n corrects this — it is called **Bessel's correction**.

Always use STDEV unless you are certain you have the entire population.

Bessel's Correction — Why $n-1$?

Proof by simulation — population = {2, 4, 6}, true variance ≈ 2.67

9 samples of size 2 — 6 unique pairs ($\times 2$ = both orderings):

Sample	$\div n$	$\div (n-1)$
(2, 2)	0	0
(2, 4) $\times 2$	1	2
(2, 6) $\times 2$	4	8
(4, 4)	0	0
(4, 6) $\times 2$	1	2
(6, 6)	0	0
Average (all 9)	1.33 ✗	2.67 ✓

- Each row is one possible draw from {2, 4, 6}
- Dividing by $n = 2 \rightarrow$ average **1.33** — always undershoots the true variance
- Dividing by $n-1 = 1 \rightarrow$ average **2.67** — matches exactly

💡 Estimating the mean "uses up" one piece of information, leaving only $n-1$ independent pieces to estimate spread — this is called a **degree of freedom**.

When Does the Mean Work Fine?

The flaw of averages is real — but the mean is not always wrong. Knowing when to trust it matters as much as knowing when not to.

Mean is appropriate when:

- The distribution is roughly symmetric (mean $\approx$ median)
- No single extreme value drives the outcome
- You need a total or balance — energy budgets, material quantities, cost projections
- The consequence of being wrong is low and reversible

Design percentile is required when:

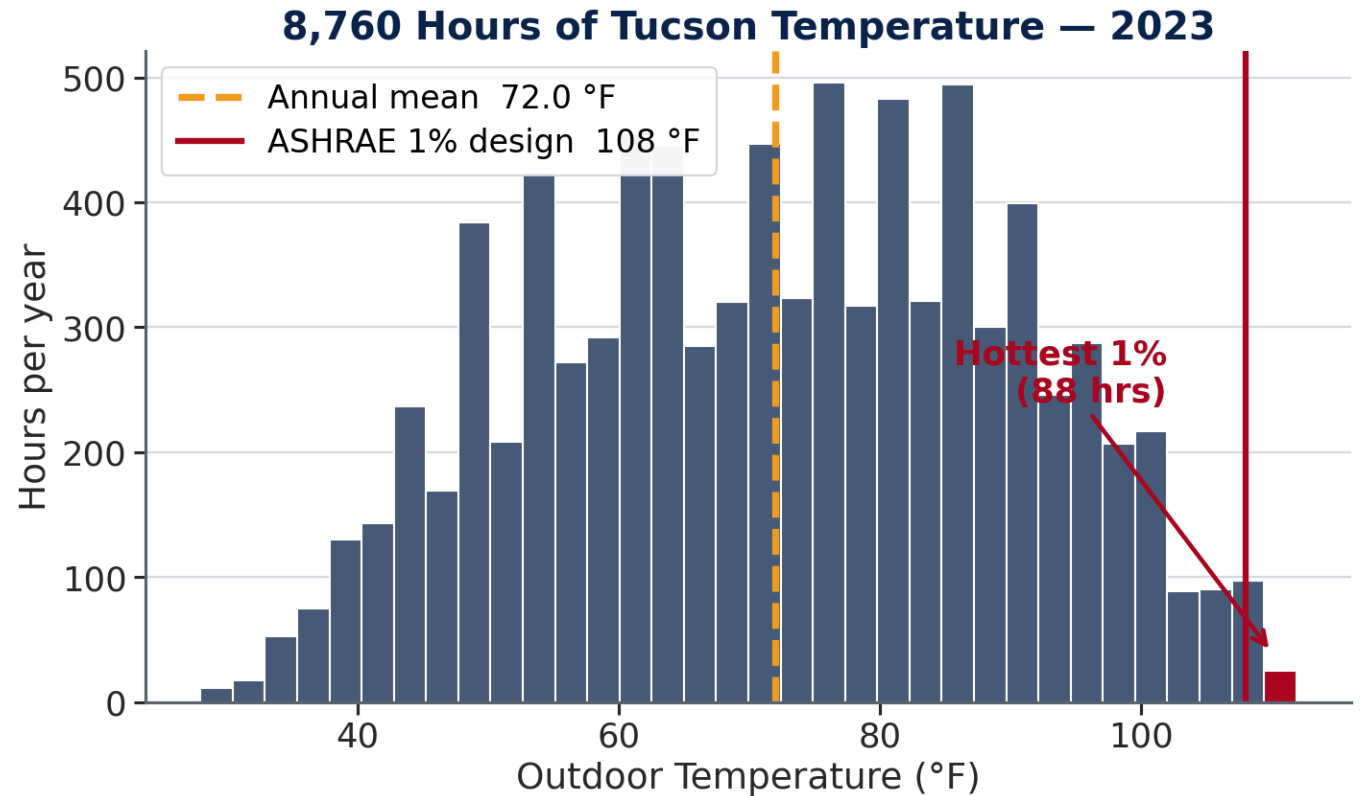
- Failure at an extreme has serious consequences
- The distribution is right-skewed (floods, gusts, peak loads)
- A system must be sized to handle the worst case it will face
- *Energy budget* → mean; *Cooling capacity* → 99th percentile

Tucson 2023 — Hours Above Threshold

How often does Tucson actually exceed design temperatures?

Threshold	Hours per year
Above 95 °F	≈ 900 hrs
Above 100 °F	≈ 429 hrs

429 hours is nearly **18 full days** of continuous operation above 100 °F — not a rare spike. A cooling system sized for the annual average (72 °F) would be undersized for months of every year.



✓ How Did You Do? — Tucson 2023

From your guesses a few slides ago

Your prediction	What the data shows
Annual average temperature	$\approx 72^{\circ}\text{F}$
Hottest recorded temperature	112°F — July 13, 2023
Hours above 100°F	≈ 429 hours (~18 continuous days worth)

The gap that matters: the mean is 72°F — comfortable, unremarkable.

The extreme is 112°F — **40°F above the mean**, and the number that sizes your HVAC system.

Most people underestimate how many hours exceed 100°F . 429 hours is nearly 6 weeks of working hours in a year — not a rare spike.

Part 3

Design Percentiles

Designing for the extreme, not the average

When Does a System Get Stressed?

Not during the average hour — during the **worst** hour.

How far into the tail? It depends on the consequence of failure:

- **HVAC comfort failure** (July heat) → discomfort, lost productivity → ASHRAE 1% (≈ 88 hrs/yr acceptable)
- **Structural failure** (windstorm) → collapse, loss of life → ASCE 7 extreme return period

The same tool — a percentile from the tail — but the depth is chosen by the engineer based on what failure means.

What Is a Percentile?

Imagine sorting all 8,760 annual temperatures from hottest to coldest.

- **Rank 1** — the single hottest hour of the year ($\approx 112^\circ\text{F}$)
- **Rank 88** — the 88th hottest hour of the year
- **Rank 4,380** — the middle value (the median, 50th percentile)
- **Rank 8,760** — the coldest hour of the year ($\approx 28^\circ\text{F}$)

A **percentile** is simply a rank expressed as a fraction of the total.

ASHRAE 1% cooling design temperature = the temperature at rank 88:

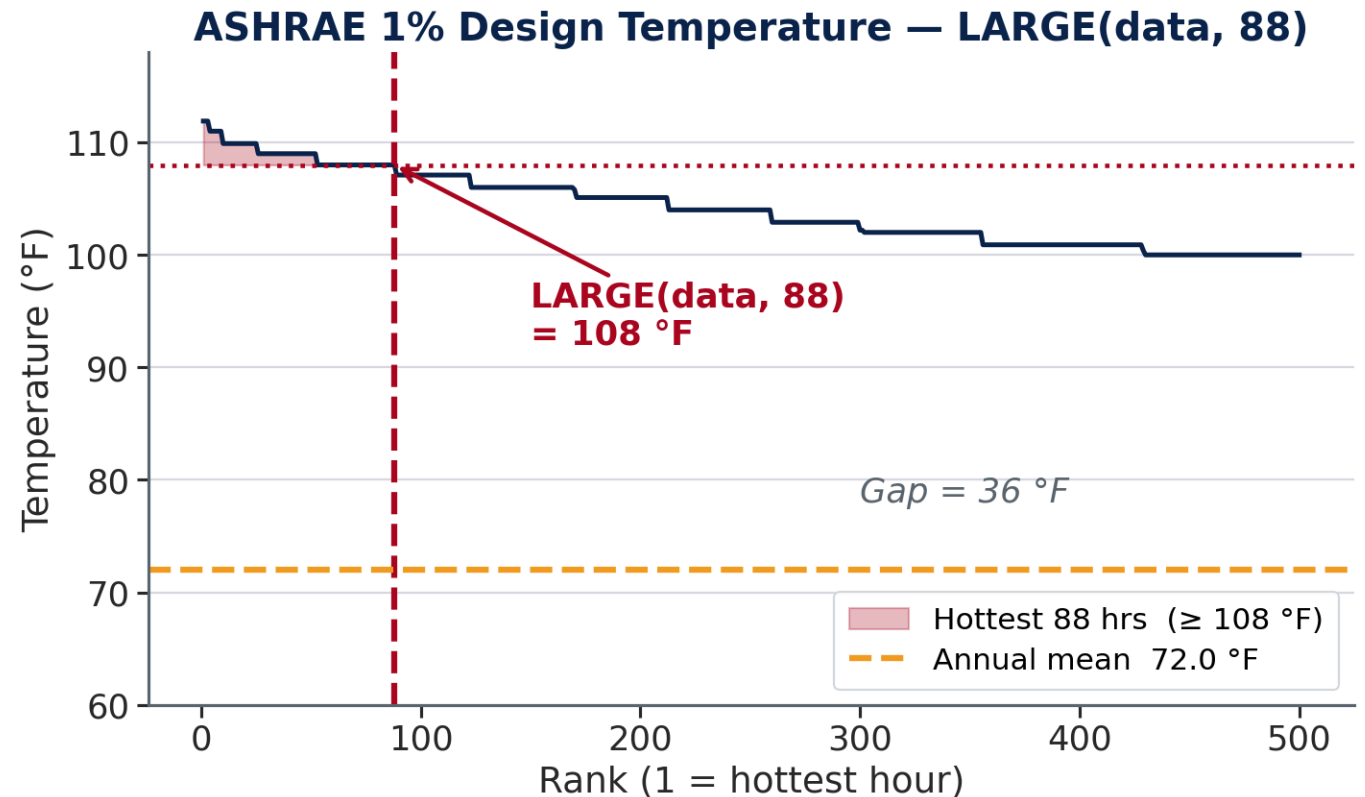
- $1\% \times 8,760 = \mathbf{88 \text{ hours}}$
- Size your cooling system for this temperature $\rightarrow$ it works for **99% of all hours**

The ASHRAE 1% Design Temperature — 108 °F

All 8,760 hours sorted from hottest to coldest. The top 88 hours (1% × 8,760) are highlighted in red.

The temperature at rank 88 is the **ASHRAE 1% design temperature: 108 °F.**

Size your cooling system to handle this temperature → it performs correctly for 99% of all annual hours.



ASHRAE = American Society of Heating, Refrigerating and Air-Conditioning Engineers · 1% design temperature = threshold exceeded in only 88 hrs/yr · the US standard for HVAC equipment sizing

Design Temperature vs. Annual Average

Statistic	Value
Annual average temperature	$\approx 72\text{ }^{\circ}\text{F}$
ASHRAE 1% design temperature	$\approx 108\text{ }^{\circ}\text{F}$
Gap	$\approx 36\text{ }^{\circ}\text{F}$

A cooling system sized for the annual average would be **undersized by 36 °F** on the hottest days.

💡 A 36 °F gap between the average and the design temperature is not just a number — it represents a dramatic undersizing of any system designed for the mean. Part 5 (Application 1) quantifies exactly how large that difference becomes in terms of required cooling capacity.

Spread — Quartiles and IQR

Quartiles are **percentiles at 25% and 75%** — the same concept as the ASHRAE design temperature, applied to the middle of the distribution instead of the tail.

Quartiles divide sorted data into four equal parts:

	What it is	Tucson 2023
Q1 (25th percentile)	75% of values are above this	≈ 58 °F
Q2 (50th percentile)	The median	≈ 73 °F
Q3 (75th percentile)	25% of values are above this	≈ 86 °F
IQR = Q3 – Q1	Span of the middle 50% of data	≈ 28 °F

Why IQR matters: unlike the range (which is pulled by the single most extreme value), IQR describes the spread of the typical half of the data — robust to outliers.

Quick Vote — January or July?

No peeking at the data · raise your hand

We know:

- **January** average ≈ 51 °F — cool, dry winter
- **July** average ≈ 94 °F — hot monsoon season

Which month has **more day-to-day temperature variability**?

- A)** January — cold fronts and warm spells create wide swings
- B)** July — the monsoon creates unpredictable conditions
- C)** About the same — high average = high variability

Part 4

Conditional Statistics

Asking smarter questions of your data

The Structure Hidden Inside the Annual Average

When you compute the annual average, you combine all 8,760 hours — January and July together.

But January and July are completely different conditions for an engineer. Combining them into one number erases that structure.

Conditional statistics answer a more specific question:

What is the average — for a specific subset of the data?

Instead of: "What is the annual average temperature?"

Ask: "What is the average temperature **in July**?"

Or: "How variable is temperature **in winter**?"

Slicing the data by a condition — month, time of day, building type, occupancy state — reveals patterns the overall average hides.

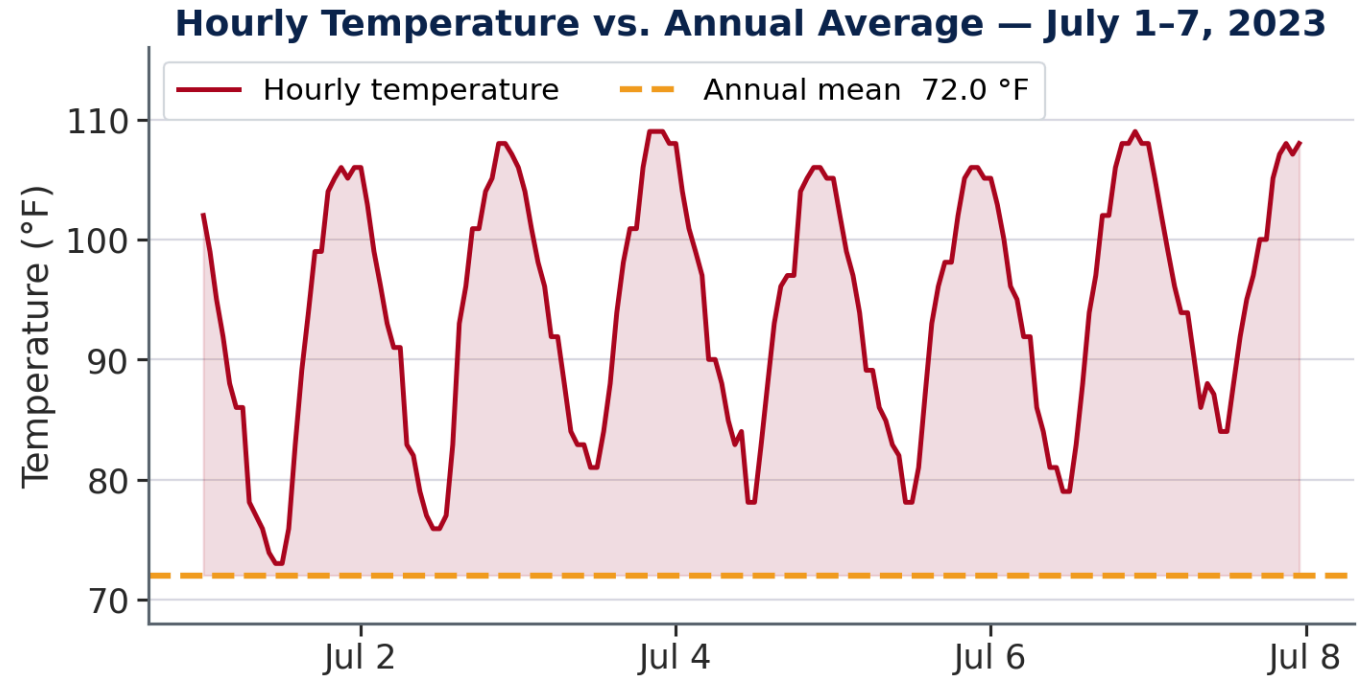
Not All Hours Are Equal

The annual average hides seasonal structure.

- January avg $\approx 51^{\circ}\text{F}$
- July avg $\approx 94^{\circ}\text{F}$
- Gap: 43°F

Every July hour is above the annual mean (gold line). The pink fill is the cooling load the average misses.

Conditional statistics ask: *what is the mean for a specific subset?*

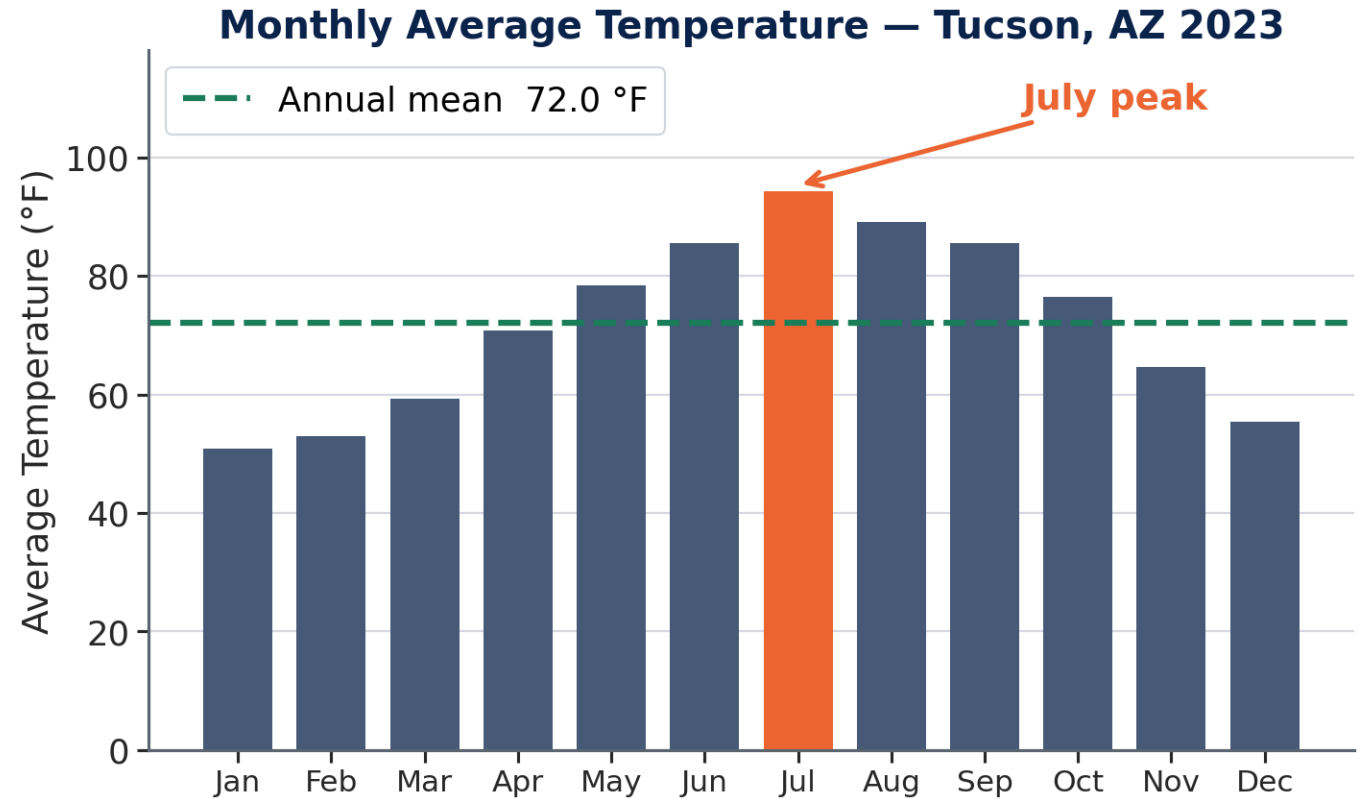


Monthly Temperatures — The Hidden Structure

Conditioning on month reveals what the annual average hides.

Month	Average temp
January	51 °F (43° below July)
July	94 °F (22° above annual mean)

5 months sit above the annual mean of 72 °F; 7 sit below. The distribution of monthly averages is itself right-skewed — the summer heat cluster pulls the mean up.



Monsoon Comfort Burden – Two Conditions

Peak heat and peak humidity do not coincide in Tucson.

Condition	Hours per year
Temp > 80 °F and RH > 50%	144 hrs
Temp > 75 °F and RH > 60%	116 hrs

Tucson's hottest afternoons stay dry. Monsoon humidity peaks in the evening, after temperatures drop — so the worst comfort conditions are warm, humid evenings, not peak-heat hours.

💡 An HVAC system sized for peak temperature alone misses the latent cooling burden. These 116–144 hours require dehumidification capacity, not just sensible cooling.

Conditional Variability — January vs. July

Month	Mean temp	Standard deviation
January	51 °F	$\sigma \approx 11.0$ °F
July	94 °F	$\sigma \approx 6.5$ °F

Cold fronts push January temperatures as low as 25 °F; warm spells reach 70 °F — all within the same month. July's monsoon locks afternoon highs in a predictable 90–105 °F band.



Quick Vote — The Answer Was A: January

Here's what the data says

Month	Average temp	Standard deviation
January	51 °F	$\sigma \approx 11.0$ °F ← more variable
July	94 °F	$\sigma \approx 6.5$ °F

Why? July's monsoon holds a predictable 90–105 °F cycle; January swings 25–70 °F between cold fronts and warm spells — wider range despite the lower mean.



Key principle: average and variability are independent. January is colder on average *and* nearly twice as variable.

Engineering implication: a Tucson heating system must cover January's 45 °F within-month range (25–70 °F) — wider than any summer month, despite July's higher average.

Part 5

Three Engineering Applications

Application 1 — HVAC Cooling (ArcE / ME)

Design question: What cooling capacity does a Tucson office building require?

Average-based (wrong):

Outdoor temp $\approx 72^\circ\text{F}$ · Indoor setpoint = 75°F

$\Delta T = 3^\circ\text{F} \rightarrow$ very small cooling load

Design-percentile approach (correct):

ASHRAE 1% design temp = 108°F

$\Delta T = 33^\circ\text{F} \rightarrow$ **11× larger load**

Size the chiller for this demand

 A chiller sized for the annual average fails on every peak summer afternoon — roughly 429 hours above 100°F per year. That is 18 full days of system failure.

Application 2 — Wind Pressure (CE)

ASCE 7-22 (*ASCE Minimum Design Loads Standard — the US code for wind, seismic, and snow loads*)

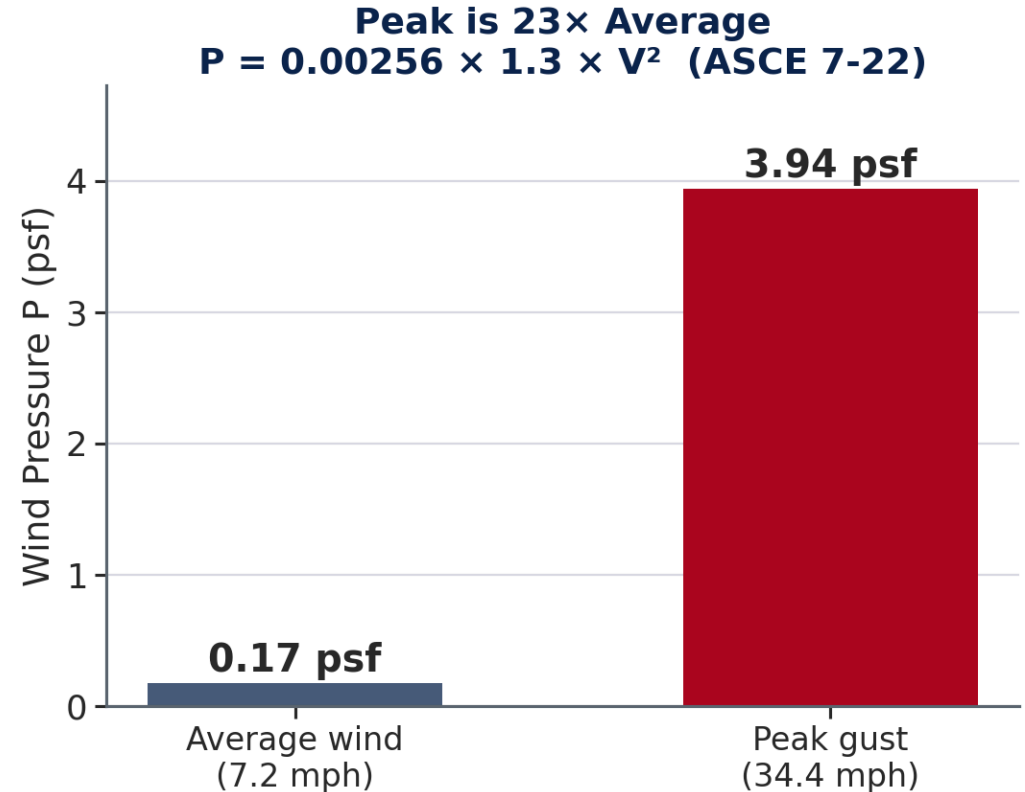
$$P = 0.00256 \times C_a \times V^2, C_a = 1.3$$

Peak gust is only **4.7× faster** — but V^2 means **23× more pressure**.

$$\frac{P_{peak}}{P_{avg}} = \left(\frac{34}{7.2} \right)^2 \approx 23$$

A billboard sized for average wind sees **23× its design load** at peak gust.

💡 Any V^2 -dependent quantity (wind, seismic, head loss) amplifies the average-vs-peak gap.



Application 3 — Latent Load / Humidity (ArcE / ME)

ASHRAE 62.1 requires HVAC systems to remove moisture from outdoor ventilation air — on top of temperature control.

As seen in the Part 4 Monsoon slide: Tucson records **144 hrs/yr** above 80 °F and 50% RH, and **116 hrs/yr** above 75 °F and 60% RH — despite its desert identity.

Mold risk rises sharply above 60% RH. These monsoon hours require **dehumidification capacity** that a temperature-only analysis misses entirely.

Design rule: size the dehumidification coil for the peak ventilation moisture load — not the annual average RH ($\approx 45\%$).

💡 A building that maintains the correct temperature but no moisture control will grow mold during every monsoon season — regardless of how well the sensible cooling system is sized.

Part 6

Lab Today · Week 2 Preview

What You Will Do in Lab — Week 1

Dataset: `Week1_Tucson_Weather_2023.csv` — 8,760 rows, 4 columns · load into Excel

Section A — mean, median, max, min, range, σ , Q1/Q3/IQR, exceedance counts, hottest month

Section B — ASHRAE 1% design temperature (B1–B2) · conditional σ by month (B3–B6) · wind pressure (B7–B11, CE) · humidity exceedance (B12–B14, ArcE)

Memo — 3 paragraphs: averages problem · design temperature · latent cooling burden

Pacing: Section A + B1–B6 + Chart 1 in class · B7–B13, Chart 2, memo at home

Lab Template Rule

Your answers are read from **specific cells in a specific file**.

⚠ Do not rename the file, change the sheet tab, move rows, or put answers in the wrong column.
Answers outside the designated cell receive zero — no manual re-grading after the deadline.

Formulas go in Col C as text — put an apostrophe before the `=`, e.g. `'=AVERAGE(B2:B8761)`.

Screenshots are not accepted.

No Deadline Extension Rule:

- After the deadline, you will not get a chance to submit or re-submit, no exceptions.

Week 2 Preview — Distributions

You computed a lot of statistics today — but statistics alone don't show **shape**.

Think about this before next class:

If you plotted all 8,760 temperature values as a histogram — bins on the X-axis, count of hours on the Y-axis — what shape would you expect?

- Symmetric? Skewed? Bimodal?
- Where would the peak(s) be?
- How does the shape connect to what you already know about Tucson's seasons?

We will build this histogram in Week 2 — and learn how to match the shape to a named probability distribution.

Week 1 Summary

The flaw of averages · designing for the mean fails when extremes drive the outcome

Descriptive statistics · mean · median · mode · range · IQR · σ · skewness · percentiles

Design percentile · ASHRAE 1% $\approx$ 108 °F for Tucson

Conditional statistics · monthly averages · conditional spread · seasonal structure hidden by annual average

Three applications · HVAC sizing · Wind pressure (V^2) · Latent monsoon load

AI connection · these tools are the foundation of every prediction model you will use or build